

Energy Procurement Optimization

Multi-Objective Linear Programming (MOLP) Approach

Decision Tree

Linear Programming

Green Energy

Executive Summary

Objective

A manufacturing plant must procure electricity every hour to meet production demand.

Determine the **optimal hourly portfolio mix** – how much electricity to procure from each source – to balance simultaneously:

- (1) **minimize total cost;**
- (2) **minimize supply risk;** and
- (3) **Minimizing CO₂ intensity.**

Context & Challenges

No single source satisfies all three sub-objectives. The optimal mix depends on prices, demand levels, and how much the company values sustainability.

Three available sources:

	Cost	Supply Reliability	CO ₂ intensity
(1) Standard Contract	Medium	Generator failure possible	Upon hourly grid mix
(2) RE Certified Contract	High	Generator failure possible	0
(3) Spot Market	Low (avg)	Higher if reported timely to operator	Upon hourly grid mix

Proposed Model

Scenario-based MOLP with:

- (1) **Decision tree** structures demand & price scenarios; and
- (2) **Multi-objective Linear Programming** solves optimal portfolio allocation in expectation across all scenarios.

Pipeline

1 – Decision Framework

Define scenarios & MOLP problem structure

2 – Data Collection

Obtain firm and electricity market data.

3 – Scenario Building

Estimate probability for decision trees from data.

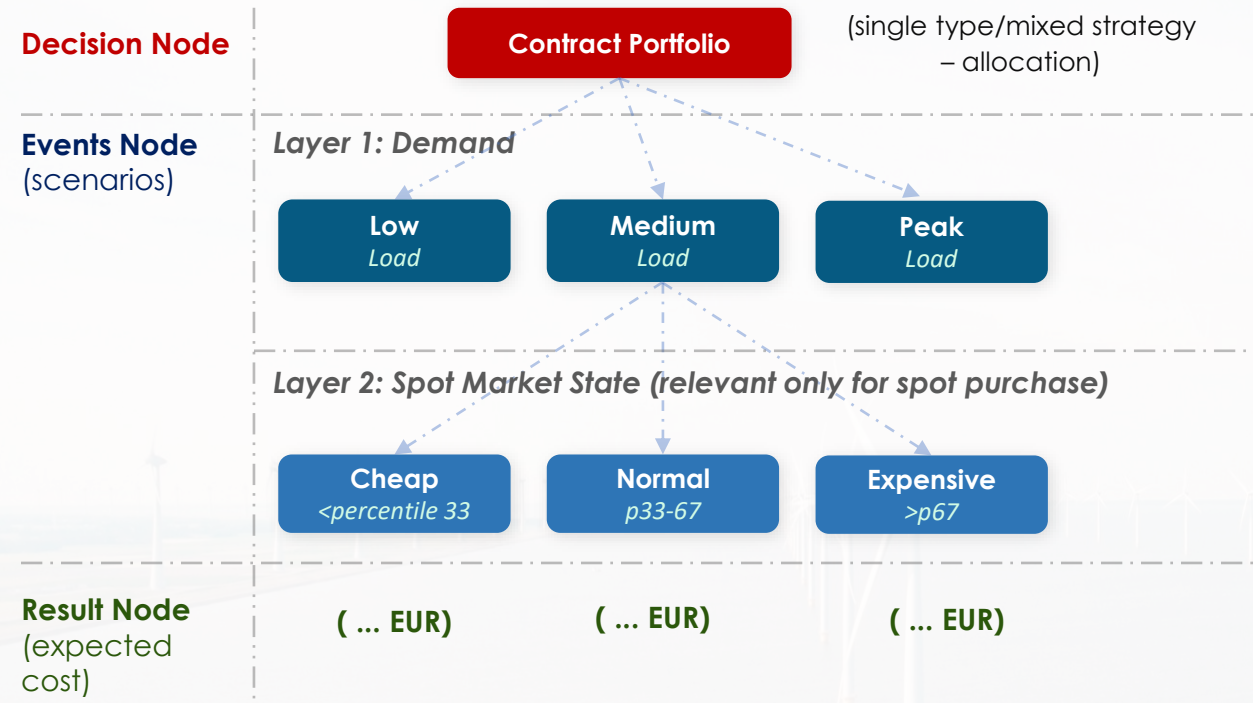
4 – Solve & Optimize

LP solver per scenario to obtain optimal portfolio.

5 – Sensitivity Analysis

Across different weights/contract premium.

Decision Tree



Scenario Structuring

- **Probabilities of each scenario are estimated from data:**
 - Demand tiers → Load_Type frequency in DAEWOO dataset
 - Price tiers → Percentile splits (P33 / P67) from ENTSO-E Finland spot prices
- Each leaf node probability = product of conditional probabilities along its path.
- **Total scenarios:**
 - If spot prices involved: up to 9 leaf nodes (3 × 3)
 - Fixed contracts only: 1 leaf node

Optimization Model – Objectives

Aggregate Objective Function

$$\min_{x_1(t), x_2(t), x_3(t)} \sum_s p(s) \cdot [w_1 \cdot \text{Cost}(s) + w_2 \cdot \text{Supply Risk}(s) + w_3 \cdot \text{CO}_2 \text{ Intensity}(s)]$$

$x_1(t)$ $x_2(t)$ $x_3(t)$ **Decision Variables (solved by LP)** – MWh bought from Standard Contract/ RE-Certified Contract/ Spot Market

w_1 w_2 w_3 **Priority Weights (set by the company)** – How much company values cost savings/ supply reliability/ green score.

s $p(s)$ **scenario s** (combination of time context, demand level, spot price state) & **probability $p(s)$** of each scenario

$$\textcircled{1} \text{ Cost} = \sum_t \sum_{i=1}^3 p_i(t) \cdot x_i(t)$$

where:

p_1 standard contract price = $E[\text{spot}] \times 1.10$
 p_2 RE contract price = $E[\text{spot}] \times 1.15$
 $p_3(t)$ spot price at hour t

Multiply price [€/MWh] $\times$ quantity [MWh] $\rightarrow$ cost [€]

Note: t = each hour in planning horizon (8,760 hrs/year)

$$\textcircled{2} \text{ Supply Risk} = \sum_t \text{short_fall}(t) \cdot [p_{\text{bal}} - p_3]$$

where:

short_fall (t) = $\max(0, \epsilon \cdot (x_1 + x_2))$: possible amount that contract generator fails to deliver
 ϵ : fraction of committed amount contract generator fails to deliver
 p_{bal} : price to buy from balancing market

Captures production loss from unmet demand

Assume baseline $\epsilon(t) = 0.02$,

$$\textcircled{3} \text{ CO}_2 \text{ Ints.} = \sum_t (x_1(t) + x_2(t)) \cdot ci(t)$$

where:

ci(t): grid carbon intensity, varies by hour

Unit: gCO₂

Optimization Model – Programming

The LP solves for $x_1(t)$, $x_2(t)$, $x_3(t)$ that **minimize the probability-weighted sum of all three objectives** across all defined scenarios.

Constraints:

	Inequality/Equation	Explanation
(1) Demand Coverage	$x_1(t) + x_2(t) + x_3(t) = D(t)$	Must cover demand at each hour
(2) Standard Contract Cap	$x_1(t) \leq \overline{C_1}$	Cannot exceed agreed contract capacity
(3) RE Contract Cap	$x_2(t) \leq \overline{C_2}$	Cannot exceed agreed contract capacity
(4) Non-negativity	$x_1(t), x_2(t), x_3(t) \geq 0$	Cannot procure negative quantities

* Note: Assume $\overline{C_1}, \overline{C_2} = 50\%, 20\%$ of average $D(t)$

Expected Outputs

01

Scenario Probability Table

Cross-tabulated probabilities for all leaf nodes, derived from data.

02

Optimal Portfolio Mix

Hourly x_1 / x_2 / x_3 allocation per scenario.

03

Cost vs Baseline

Expected savings vs naive always-spot strategy, quantified in €/year.

04

Trade-off Frontier

How optimal cost and green score shift as w_1 , w_2 , w_3 weights change.

05

Contract Premium Sensitivity

Optimal mix at 5% / 10% / 15% premium – value of contract stability.

06

Supply Reliability Analysis

Impact of contract generators' failure to deliver on cost/shortfall